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17 gr bakır sülfat tuzu
10 ml H_2SO_4
90 ml saf su
} 100 ml çözelti

6.558 g → katot plakanın ağırlığı — ilk (paslanmaz çelik)

EXPERIMENT 7: REDUCTION ELECTROLYSIS

1. AIM OF THE EXPERIMENT

In this experiment; It is aimed to learn the electrolysis phenomenon, to learn the refining (purification) and reduction (recovery) processes of copper and to obtain electrolytic copper by using copper in solution.

kaplama → saflaştırma

Anotla katotu bağlarken akımı neye göre belirleriz.

2. THEORETICAL INFORMATION

Electrometallurgy

Akım yoğunluğu → 15mA - 30mA arası uygulursanız çoğu zaman ideal → safsızlıkların daha az olduğu bir kaplama olur.

It is called "Electrometallurgy" to produce metals in all kinds of raw materials containing ore or metal by using electrical energy. In reality, electrometallurgy is an application of electrochemistry. Here, electrochemistry methods are applied to metals.

Electrolysis

Akım yoğunluğu hesabı = akım yoğunluğu = $\frac{i}{A} = \frac{i}{30cm^2} \times 0.015\ A$

$i = 0.45\ A$ kullanacağım akım miktarı

It is the phenomenon of the liberation of metal ions at the cathode and non-metallic ions at the anode as a result of chemical decomposition by passing an electric current through aqueous or molten electrolytes. Ions are discharged at the electrodes; It can be liberated, collected as an element, and react with the electrode.

Akım → akım yoğunluğu ve alana göre seçilir.

Electrode: Conductor that conducts electric current to liquid, solid and gas phases. In an electrochemical sense, the electrode is the system that combines the electrode conductor and the ionic conductor.

Electrolyte: It is the environment that contains (+) and (-) charged free ions. It is the liquid that conducts the electric current used in electrolysis. Molten salts and solutions of acids, bases and salts are used as electrolytes.

6dk süre ile 0.45 A akımla bu bileşiği yapıcaz $M = 63.5$ $z = 2$ → $M = \frac{MIt}{zF}$

Electrolysis cell: Cells in which electrical work is converted into chemical work by applying external voltage.

0.064 g biriktirme yapmayı hedefliyoruz.

Generator: It is a direct current source that provides electrical energy in the external circuit. The electrode connected to the positive pole of the generator is the anode, and the electrode connected to the negative pole is the cathode.

In electrolysis; cations (+ charged ions) are reduced to the cathode, and anions (- charged ions) are oxidized to the anode. If there is more than one type of cation in the electrolysis vessel, the one with the greatest reduction potential (potential) is reduced first. Then the reduction continues sequentially. If there is more than one type of anion in the electrolysis vessel, the anion with the greatest oxidation tendency is collected at the anode first. The less active anion is first released at the anode, and the less active cation is first released at the cathode. In electrolysis, compounds can be separated into their elements. If a voltage higher than the battery voltage is applied to the electrochemical cell, the reaction in the battery is reversed. Electrolysis occurs. This phenomenon is called charging the battery.

The electrolysis process is carried out in a cell called an electrolysis vessel or tank. This cell consists of two electrodes immersed in a compound that dissolves into positively and negatively

Anot asidik olmaya katot bazik " meyillidir

6dk → 360s yaptık. → $\frac{63.5 \times 0.45 \times 360}{2 \times 96485} = m$

m = 0.053 g bekliyoruz.

birikmeden sonra
saf su ile yıkıyoruz.
sonra kurutmaya attık.
Sonra tartıldı
4.617g → 0.059 g biriktirmişik

charged ions, and these electrodes are arranged so that they do not touch each other (usually 5- 20 cm between the two electrodes). In order to carry out the electrolysis process, these electrodes are connected to a direct current source and the voltage (electric field) between the electrodes moves the ions towards the oppositely charged electrode (pole). Therefore, (+) charged ions go to the cathode, while (-) charged ions flow towards the anode. Atoms or molecules that balance their charge at the opposite pole precipitate on the electrode or enter into new reactions with the molecules in the electrolyte.

Rafinasyon da aynı cihazla yapıyoruz. Biriktirme yaptığım plakayı takıp karşısına da saf olduğunu bildiğim bakır elektrodu takıyoruz

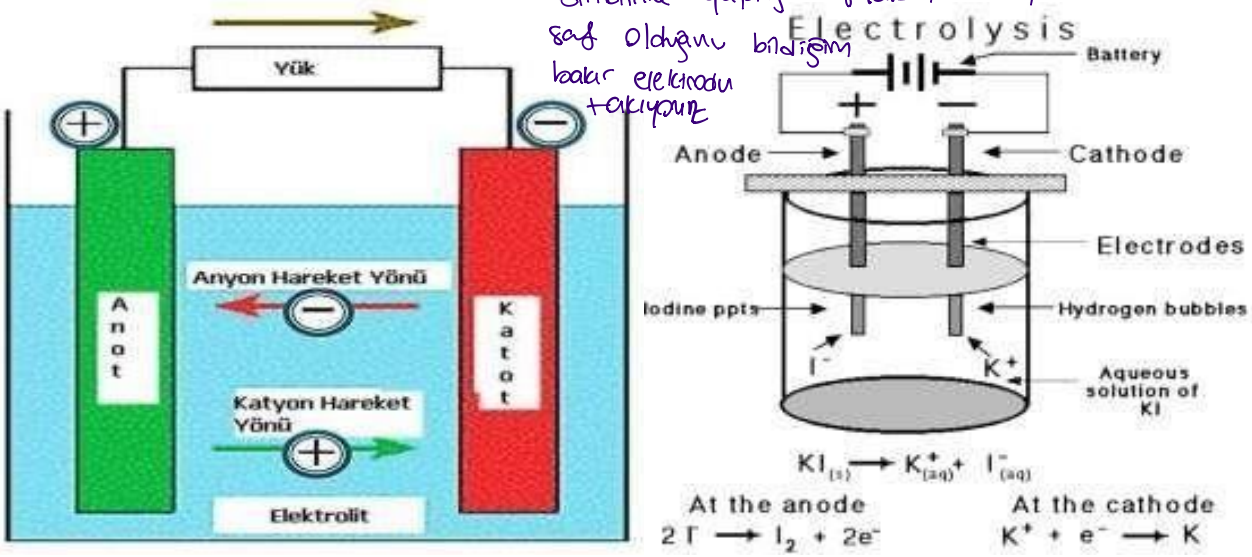


Figure 1. Electrolysis Method

Bakır elektrodun kütlesi → 20.104 gram başlangıçta

Electrolysis of copper (Cu) in industry is used for two purposes. One of them is the purification of copper, which is carried out in order to collect the impure copper metal from the impurities in it and collect it on the cathode; The other is the copper recovery process to recover the copper metal from aqueous solutions.

AYM 6 dk rafinasyon yapcam →

Katodumuz anot oldu

Application Areas of Electrolysis

Electrolysis processes, as mentioned earlier, are used in the field of metallurgy by electrolysis, in the preparation (method using insoluble anode-recovery electrolysis) or refining (method using soluble anode-purification electrolysis) of metals. It is also a method of protecting against corrosion by an electrolytic metal deposit and coating metals with a metal deposit (for example, nickel plating, zinc plating, cadmium plating, chrome plating, silver or gold plating). Pure hydrogen is also obtained, in particular, by the electrolysis of water. Application areas include gas production (chlorine), obtaining anode layers with protective oxides on metal (the process of anodizing aluminum by means of alumina), electropolishing, degreasing of metals as cathode or anode. Electrolysis also allows the measurement of current intensities, especially current amounts in voltmeters. Treatment electrolysis, which is based on the separation of organic tissues with the help of continuous current, is also used in medical applications such as the destruction of nerve endings (neurons), hardened tumors, polyps in the nostrils, and the treatment of urethra or esophagus narrowing.

Refining (Purification) Electrolysis of Copper

Rafinasyondan sonraki kütle

20.189 Bakırda biriken

0.088 artış

The refining electrolysis of copper is an electrolysis process that enables to obtain refined copper of sufficient purity, and this process is carried out with soluble copper anodes. Copper sulfate ($CuSO_4$) and H_2SO_4 solution are used as electrolytes. One of the copper electrodes immersed in the solution is pure copper, while the other is impure copper. As the (+) charged anode, copper (blister copper) electrodes obtained by purification at high temperatures (98-99% purity) are used. These electrodes contain O, S, Au, Ag, Pt and various amounts of As, Sb, Bi, Sn, Se, Te, Pb, Zn, Fe as impurities in their composition. As the (-) charged cathode, titanium plates coated with electrolytic copper or copper plates (99.5% purity) coated with oil to prevent sticking are used. While electrons flow from the anode to the cathode out of the electrolyte; They flow from the cathode to the anode in the electrolyte. Under suitable conditions, the anode oxidizes and goes into solution and is reduced at the cathode. Some of the other impurities in the anode do not dissolve, they break off from the anode and collect at the bottom of the cell. This residual material is called "anode mud". Some impurities dissolve into the electrolyte. Since the anode and cathode are of the same composition, the separation voltage is theoretically zero and the cell voltage is slightly above the electrolyte resistance.

Electrolytic refining is done for two reasons. While the main purpose is to get rid of impurities that cause a decrease in conductivity, noble metals and semi-metallic metals (Se,Te) that pass into the anode sludge during the refining process add value to the business as they cover the cost in the refinery.

When current is applied between the anode and the cathode, the following occurs in the $CuSO_4$-H_2SO_4-H_2O type electrolyte:

The operating conditions of classical copper purification electrolysis are 0.2-0.35 V cell voltage. The electrolysis conditions are as follows:

- 35-50 g/l Cu and 140-220 g/l H_2SO_4 in electrolyte
- Temperature 50-65 ºC
- Cathode current density 150-250 A/m2 (15-25 mA/cm2)

Reactions occurring in the purification electrolysis of copper;

Anode reaction: $Cu^0 \rightarrow Cu^{+2} + 2e^-$ (Here, copper dissolves as Cu+2 ions as a result of anodic oxidation.)

Dissolved (+) charged copper ions (Cu^{+2}) reach the cathode surface.

Cathode reaction: $Cu^{+2} + 2e^- \rightarrow Cu^0$ (Here Cu^{+2} copper ions are reduced to metal state by recombining with electrons transferred from the anode.)

0,045 A 4.526 g başlangıç kütlesi

t = 10 dk x 60 = 600 s

$$\frac{0{,}145 \times 600 \times 63{,}5}{2 \times 96485} = 0{,}0888\ g$$ toplam

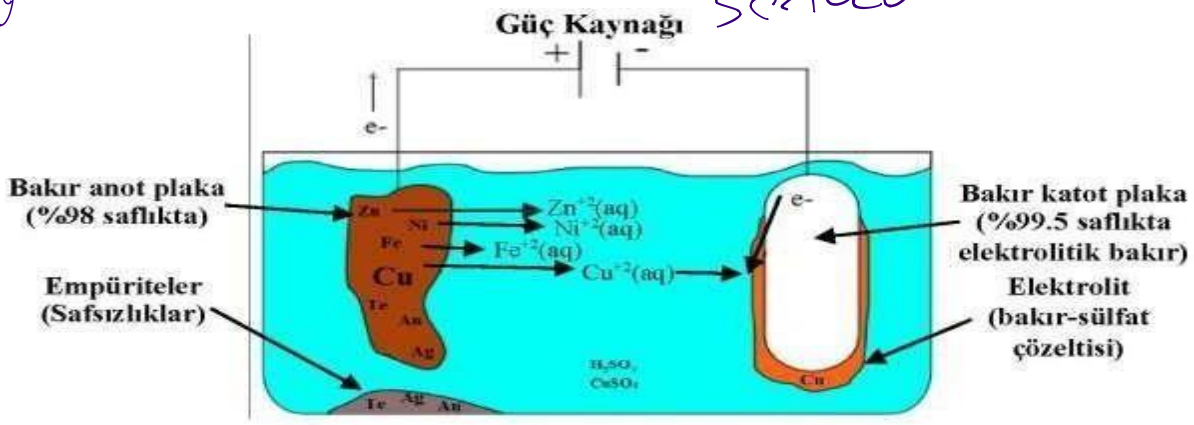


Figure 2. Purification electrolysis of copper

Among these options, it is seen that there is no energy consumption in copper refining electrolysis. However, since the anodic and cathodic reactions are kinetic impediments, excessive voltages must be applied, and the voltage drop due to the resistors at the junctions must also be considered.

Reduction (Reduction) Electrolysis of Copper

Recovery electrolysis of copper is an electrolysis process that provides copper from solutions, and insoluble anode and cathode electrodes are used during this process. While a hard lead alloy containing 4-6% Sb is used as the anode, stainless steel and recently titanium plates are used as the cathode. For the reduction process, a temperature of 40-60 ºC and a current density of 70- 150 A/m2 are selected. During electrolysis, (-) charged ions in the solution go to the anode, and (+) charged ions go to the cathode. Anions are oxidized at the anode surface, while cations are reduced at the cathode surface. Thermodynamically, copper reduction electrolysis takes place at 0.89 V.

Reactions during reduction electrolysis of copper; $CuSO_4 + H_2O \rightarrow Cu + H_2SO_4 + ½O_2$
Anode reaction: $SO_4^{-2} + H+ + 2e- \rightarrow H2SO_4 + ½O$

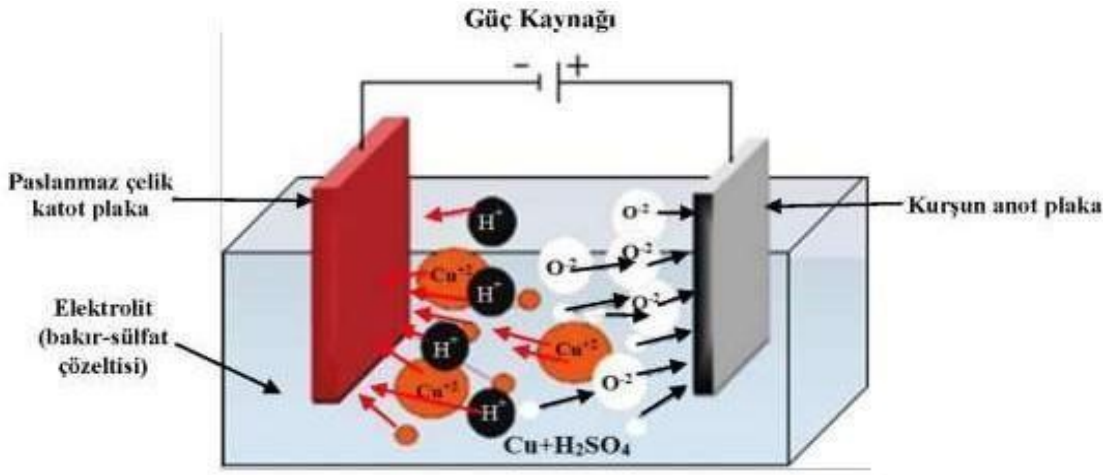


Figure 3. Recovery electrolysis of copper

One of the differences between the purification and recovery electrolysis of copper is that the 'cathode' reactions are the same but the 'anode' reactions are different. In addition, in the electrolysis

of copper purification, metal ions dissolved from the anode are reduced at the cathode; In the recovery electrolysis of copper (as an insoluble lead alloy containing 4-6% Sb is used), the anode reaction takes place by oxidation of an anion or molecule in the electrolyte.

Electrolysis Conditions and Additives

Colloid is a homogeneous mixture. This homogeneous mixture contains gelatin and thiourea. Gelatin provides homogeneous and uniform bonding of copper to the cathode surface and creates a fine crystalline hard copper structure. Thiourea is used to eliminate the side effects of gelatin. Another task of the thiourea is to isolate the spikes and pits on the cathode surface. Another additive, mersolate, acts as an antioxidant and delays sulfation on the cathode surface. The purpose of adding HCl to the homogeneous mixture is to precipitate silver ions (Ag+) in the electrolyte as silver chloride (AgCl).

Table 1. Electrolysis conditions and additives

Parameters	Conditions
The copper concentration in the electrolyte	35-50 gr/l
H_2SO_4 concentration in the electrolyte	140-220 gr/l
Temperature	50-65 ºC
Cathode Current Density	150-250 A/m^2
Cell Voltage	0,2-0,35 V
Circulation Rate	0,02 m^3/dak.
Colloids Gelatine	0,0001-0,001 kg/m^3
Thiourea	0,0001-0,001 kg/m^3
HCl or NaCl	0,035 kg/m^3

3. CALCULATIONS

In an electrolysis circuit, the amount of product that should be theoretically obtained at the end of the process by the passage of the 'I' current for 't' time is calculated by the Faraday equation:

$MT = (I . t . M_a) / (Z . F)$ * ;

In this equation,

MT = The amount of product obtained (accumulated at the cathode),

I = The current intensity (A) passed through the circuit,

t = Time (sec),

M_a = Atomic weight of the substance deposited at the cathode (for Cu: 63 gr/mol), Z = Valence of the substance deposited at the cathode in the compound (for Cu: 2),

F = Faraday's constant (96500 Coulomb).

In electrolysis processes, the current density is calculated by the ratio of the current flowing through the circuit to the area of the cathode electrode immersed in the electrolyte:

Cathode Current Density =
Current Through Circuit / Area of the Cathode Remaining in Electrolysis

Since some of the current flowing through the circuit is used to overcome the resistances during the experiments, the product obtained by electrolysis is less than the theoretical product. As a result of multiplying the ratio of the obtained product (Mg) to the theoretical product (MT) by 100, **** β = [(MG / MT)×100], % current efficiency (efficiency)** is obtained. If the amount of material collected at the cathode or separated from the anode is equal to the theoretical value calculated by the Faraday equation, it is understood that the current efficiency is 100%, and if not, it is below 100%. In addition, the value obtained as a result of **[(MT-MG)/MT)×100]** gives the **experimental % error margin.**

Case Study;

a- Refining Electrolysis

Before starting the experiment, the electrolyte liquid was prepared using the data in Table 1. The prepared electrolyte was placed in the beaker. Then the cathode plate was cleaned with alcohol and dried and its weight was measured on a precision balance. The temperature of the prepared electrolyte was increased to 65 ^{0}C and meanwhile magnetic stirrer was started. The current density was calculated by measuring the surface area of the cathode plate in the electrolyte. The prepared anode and cathode plates were connected to the power source and the current was given according to the calculated current density. After the electrolysis process took 15 minutes, the power supply was turned off just in time. The cathode plate was removed from the electrolyte, cleaned and dried with alcohol, and its final weight was measured on a precision balance. Then, the theoretical copper amount to be obtained during the experiment was calculated using the (*) equation. The current efficiency was calculated using the equation (**) to calculate the electrolysis efficiency.

b- Reduction Electrolysis

The electrolyte used in refining electrolysis was used in reduction electrolysis. Lead plate was used as the anode plate and steel plate was used as the cathode plate. After the cathode plate was cleaned with alcohol and dried, its weight was measured on a precision balance. Then, the surface dimensions of the part that will remain in the electrolyte were measured and connected to the power supply with the anode positive and the cathode negative. The voltage was increased by keeping the current applied to the refining electrolysis constant. After the electrolysis process took 15 minutes, the power supply was turned off just in time. After the cathode plate was removed, cleaned with alcohol and dried, the final weight was weighed. Then, the amount of copper to be collected during the electrolysis was calculated according to the (*) equation. To calculate the electrolysis efficiency, the current efficiency was calculated using the equation (**)

c- Example Interpretation

The yield (99.2%) we obtained as a result of refining electrolysis showed that the experiment was carried out correctly. However, in reduction electrolysis, more copper was collected than the

theoretically required amount. The reasons for this may be: the current density cannot be adjusted exactly or the anode and cathode plates are closer to each other than they should be.

The copper mined in the refining electrolysis was obtained by dissolving the fire-refined anode plate with a purity of 98-99%. On the other hand, in reduction electrolysis, the copper in the electrolyte is recovered. One of the differences between refining electrolysis and reduction electrolysis is the anode plate used. Lead plate was used in reduction electrolysis and there was no dissolution of the used plate.

The reason for using a lead plate is that the lead remains on the anode surface directly forming insoluble PbSO4. If the anode copper contains too much lead, the resulting PbSO4 completely covers the surface and causes the anode to passivation.

Another difference is the anode reactions that occur. While metal ions dissolved from the anode are reduced at the cathode in refining electrolysis, since insoluble anodes (usually lead alloys containing 4 – 6% Sb) are used in reduction electrolysis, the anode reaction must be achieved by oxidation of an anion or molecule in the electrolyte. The oxidation process is a task undertaken by the OH- ion, which is formed as a result of the dissociation of water in aqueous solutions. The refining and reduction electrolysis anode reactions are given below.

Although lead is a basic metal; Explain the reason and/or reasons why copper is used as an anode material in reduction electrolysis, accompanied by related reactions.

When we look at the EMF series, we can see that lead is more basic than hydrogen, and copper is more noble. With this simple approach, we can say that lead will dissolve if it is used as an anode during copper reduction electrolysis. However, in lead and copper reduction electrolysis, PbO and PbO formed by the breakdown of water, and this PbO makes the H2SO4 solution and PbSO4 compound. These compounds coat the lead anode surface and form a stable anode surface. As a result, lead is insoluble and is therefore used as an anode material.

$2H_2O \rightarrow O_2 + 4H^+ + 4e^-$ (Anode Reaction), $Pb + \frac{1}{2} O_2 \rightarrow PbO$,
$PbO + H_2SO4 \rightarrow PbSO_4 + H_2O$

4. MATERIALS AND DEVICES USED IN THE EXPERIMENT

- Electrolysis cell,
- Direct current power supply (voltmeter, ammeter and connection cables),
- Copper-sulphate and sulfuric acid solution (35-50 g/L Cu-containing solution in Electrolyte, 140-220 g/L H2SO4), alcohol (for cleaning), gelatin powder and thiourea (0.001 g) as additives
- Precision scales,
- Beakers,

- Cathode electrolytic copper and anode blister copper (purified at high temperatures) plates (for purification electrolysis),
- Cathode stainless steel and anode lead plates (for recovery electrolysis),
- Heater, magnetic stirrer, caliper. Duties of additives:

Gelatin: It creates a fine crystalline hard copper structure by ensuring that the copper deposited on the cathode is properly and homogeneously bonded to the surface.

Thiourea: Eliminates the side effects of gelatin and isolates the spikes and pits on the surface of the cathode material.

5. CONSTRUCTION OF THE EXPERIMENT

The electrolyte solution is prepared by weighing the specified amounts of copper-sulphate ($CuSO_4$) and H_2SO_4 and put into the beaker. Specified amounts of gelatin powder and thiourea are added to the solution and mixed. The material from which the metal will be collected (cathode plate) is cleaned with alcohol and dried, and the result is recorded by weighing it on a precision scale. The anode and cathode electrodes are placed in the electrolyte solution. The connection cable of the anode plate to the (+) pole of the DC power supply; The connection cable of the cathode plate is attached to the (-) pole. Considering 150 A/m^2 in both copper purification and recovery electrolysis experiments as current density, the area of the part of the cathode plate immersed in the electrolyte is calculated, and after determining the amount of current to be switched, the DC power supply is operated and the circuit is energized. The electrolysis process continues for 15 minutes and the power supply is turned off according to the exact time, and the current supplied to the circuit is cut off. The cathode plate is cleaned with alcohol as in the beginning, dried and its final weight is re-weighed. The amount of material collected at the cathode is determined and compared with the amount that should theoretically accumulate, the current efficiency and experimental error margin are calculated with the help of the equations given above. The execution of both experiments is carried out in the same way.

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